BELTON

CONVEYORS LTD

BX-12

Packaging Conveyor

Operation & Maintenance Manual

This manual applies to the BX-12 conveyor only. It does not apply to other Belton conveyors, which are supplied with their own manuals.

Document BC-OM-012, Issue B, 1 October 2024

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SECTION 1

About this manual

1.1 This manual covers installation, commissioning, adjustment, maintenance and fault finding for the Belton BX-12 packaging conveyor with the BCU-2 control unit. Keep it with the conveyor. Further copies can be downloaded from the customer area of our website, quoting the serial number on the rating plate at the drive end.

1.2 In this manual, left and right are as seen standing at the tail (infeed) end, looking along the conveyor towards the drive (discharge) end. Please use the same convention when you call Technical Support; it saves a great deal of confusion.

1.3 Changes in Issue B: Section 6, belt tensioning by elongation replaces the old deflection method. Section 10, fault code C-06 added for BCU-2 firmware 2.2 and later. Section 11, parts list updated.

SECTION 2

Safety

2.1 The BX-12 must be installed and used in accordance with the Provision and Use of Work Equipment Regulations 1998 (PUWER) or the equivalent rules in the country of use. Only trained persons may adjust or maintain the conveyor.

2.2 Before any work other than the running adjustments in Sections 6 and 7, isolate the conveyor at the lockable isolator on the BCU-2 and fit your personal padlock. The emergency stop is not an isolator.

> **WARNING**
> The in-running nips at the drive and tail pulleys can draw in fingers, gloves and clothing in a fraction of a second. Never run the conveyor with a guard removed.

2.3 Belt tracking has to be adjusted with the belt running. The take-up adjusting screws are outside the guarded area and are reached through the two access holes in the tail guard. Do not remove the tail guard to make adjustments.

2.4 **Lifting**
Lift conveyor sections only with slings around the side frames at the marked lifting points, never by the drive unit, the BCU-2 or the guards. A 6 m BX-12 weighs about 210 kg.

2.5 **Cleaning**
In food areas the conveyor is usually washed down every shift. Isolate it first. Keep high-pressure jets away from the BCU-2, the motor terminal box, the sensors and all bearings. Use only cleaning chemicals with a pH between 4 and 10 that are approved for polyurethane; chlorinated cleaners above 200 ppm make the belt brittle within months.

2.6 **Noise**
The airborne noise of the BX-12 is below 70 dB(A) at 1 m. A noticeable increase usually means a seized idler or a bearing on its way out. Find it before it fails.

SECTION 3

Specification

Item	BX-12
Belt	600 mm wide, PU food grade, endless (factory joined), part 77-1012
Conveyor length	2 m to 12 m centres, in 0.5 m steps (see rating plate)
Belt speed	0.1 to 0.6 m/s, set at BCU-2 (inverter)
Maximum load	35 kg per metre, evenly distributed; 150 kg in total
Drive	End drive at discharge, 1.1 kW geared motor, IP65
Drive pulley	160 mm, crowned, lagged; fixed and squared at the factory
Tail pulley	120 mm, crowned, with screw take-ups both sides
Belt support	Slider bed (top run); return idlers 77-0410 at 1 m pitch
Sensors	Belt edge sensors S3 (left) and S4 (right) at the tail; speed sensor S5 on tail pulley
Frame	Stainless steel 304, folded side frames
Control	BCU-2 control unit, 400 V 3-phase 50 Hz supply

3.1 Rating plate

The rating plate on the drive-end side frame gives the serial number, the conveyor length between pulley centres, the year of manufacture and the BCU-2 firmware fitted at despatch. Serial numbers begin BX12- followed by the year and a sequence number, for example BX12-240077.

SECTION 4

Installation and commissioning

4.1 Levelling

Set the conveyor on its legs and level it along its length and across the belt with a spirit level, using the adjustable feet. Use packers under the feet only on soft floors.

4.2 Squareness

A frame that is out of square will make the belt track to one side however carefully it is adjusted. Measure the two diagonals of the frame between the centres of the drive and tail pulley bearings. They must agree within 3 mm. Where sections are bolted together, check the joint plates are tight before measuring.

4.3 Commissioning

Before first start: check that all guards are fitted, the belt is tensioned (Section 6), the edge sensors are 5 mm from the belt edge, and the direction of travel matches the arrows on the side frame. Run the belt empty for 30 minutes and adjust the tracking (Section 7). Complete the commissioning record at the back of this manual and return a copy to Belton within 30 days: it starts the warranty (Section 12).

4.4 Electrical connection

The BCU-2 needs a 400 V three-phase supply with earth, protected by a 10 A type C circuit breaker. Cable entries are at the bottom of the enclosure; keep them there, since water runs down cables. The motor and sensor cables are factory-wired and plugged; do not cut them to length.

4.5 **Transfers**

Where the BX-12 receives product from, or passes it to, other equipment, keep the transfer gap below half the length of the smallest pack, and set the belt surface level with, or slightly below, the feeding surface. A belt that sits higher than the infeed catches the leading edge of each pack and pushes it to one side, which looks like a tracking problem but is not.

4.6 **Edge sensors**

Set the gap between each belt edge sensor and the belt edge to 5 mm with the belt running centrally. Sensors set closer than this trip C-02 on normal belt wander.

SECTION 5

The control unit

5.1 The BCU-2 has a two-line display, START and STOP push buttons, a speed potentiometer and a RESET button. Faults are shown as a C-code with a short text (Section 10). The last 50 faults are stored with date and time under MENU > LOG.

5.2 The edge sensors S3 and S4 stop the conveyor if the belt edge stays over a sensor for 2 seconds (fault C-02). They are a protection, not a tracking aid: a belt that keeps tripping C-02 needs tracking, not a sensor adjustment.

5.3 User parameters are under MENU > SETTINGS and are protected by the four-digit code supplied with the conveyor. Factory settings are restored with MENU > SETTINGS > DEFAULTS.

Parameter	Function	Range	Factory setting
U01	Belt speed	0.10 - 0.60 m/s	0.30 m/s
U02	Acceleration ramp	0.5 - 10 s	2 s
U03	Deceleration ramp	0.5 - 10 s	1 s
U04	Jam timer (C-05)	1 - 60 s	8 s
U05	Edge sensor delay (C-02)	1 - 5 s	2 s
U06	Slip threshold (C-03)	2 - 15 %	5 %
U07	Run-on after infeed stops	0 - 120 s	30 s
U08	Restart after power loss	Manual / Auto	Manual

5.4 Increasing U05 to stop nuisance C-02 trips only lets the belt run further off before it stops. Leave it at 2 s and track the belt.

SECTION 6

Belt tensioning

6.1 The belt needs just enough tension to drive without slipping. Too little and the belt slips on the drive pulley (fault C-03). Too much and it overloads the pulley bearings, stretches the belt and makes tracking harder, not easier.

6.2 **Elongation method**

Belton specifies tension as elongation of the belt: 0.3 - 0.5 % of its relaxed length. Measure it like this:

a) Isolate the conveyor. Slacken both take-up screws until the belt is just slack.

b) On the top run near the middle of the conveyor, mark two fine lines across the belt exactly 1000

mm apart.

c) Tighten the left and right take-up screws equally, counting the turns, until the marks are 1003 mm to 1005 mm apart (0.3 - 0.5 %).

d) Tighten the take-up lock nuts to 35 Nm.

e) Run the belt and check the tracking (Section 7).

6.3 New belts

Set a new belt to 0.3 %. A new belt relaxes during running-in. Check the marks again after 8 hours and 50 hours of running and re-tension to 0.3 - 0.5 % if needed. Always turn both screws by the same amount, or you will change the tracking as well as the tension.

SECTION 7

Belt tracking

7.1 How tracking works

A belt moves towards the end of a pulley or roller that it touches first. The crowned drive and tail pulleys pull the belt towards their centre, but only if the frame is square, the pulleys are square to the frame and every roller the belt touches is free and square. Tracking on the BX-12 is adjusted at the tail pulley only. Do not move the drive pulley; it is squared at the factory.

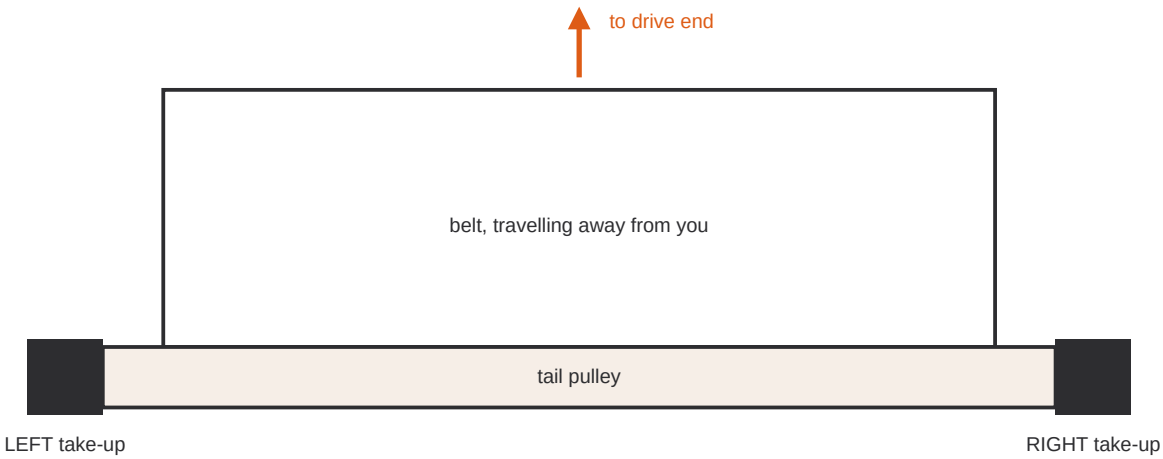

Figure 7-1 Tail end seen from the infeed. Belt running to the left: tighten the LEFT take-up (or slacken the right). Belt running to the right: tighten the RIGHT take-up.

7.2 Procedure

Run the belt empty at the normal speed. Watch the belt at the tail pulley for at least three full revolutions of the belt before deciding which way it is going.

a) If the belt runs off to the left, tighten the left-hand take-up screw a quarter of a turn, or slacken the right-hand screw a quarter of a turn. If it runs to the right, do the opposite.

b) Wait for three revolutions of the belt. The belt reacts slowly; impatience is the commonest cause of over-correction.

c) Repeat in quarter turns until the belt runs centrally at the tail and at the drive.

d) Load the belt with typical product and check again.

e) Re-check the tension (Section 6). Tracking adjustments change it slightly.

> **NOTE**
> Make small adjustments only. If you need more than two full turns of difference between the left and right take-up screws, the cause is somewhere else and the adjustment is only hiding it.

7.3 **If the belt will not track**

Belts that track badly after a belt change usually have one of these causes. Check them in this order:

- **a)** Take-ups not equal. Compare the left and right screw projections with a rule.
- **b)** Frame out of square (4.2), often after the conveyor has been moved or a leg has been knocked.
- **c)** Product residue or wear on the pulley faces. Clean with warm water; never use solvent on the lagging.
- **d)** Return idlers. Spin each idler by hand. A tight, seized or bent idler, or one not square to the frame, steers the belt. Replace damaged idlers (77-0410).
- **e)** The belt itself. Lay the old and new belts flat and compare the edges with a straight edge. A belt that is not straight (cambered) will never track; return it to Belton.

7.4 If the belt still tracks to one side after these checks, stop and call Belton Technical Support with the serial number, the direction of tracking, and the turns already applied at each take-up.

SECTION 8

Belt replacement

8.1 The BX-12 belt is endless and is replaced from the right-hand side, which has removable bed plates and a cantilevered return frame. Allow one hour for two people on a 6 m conveyor.

- **a)** Isolate and lock off the BCU-2.
- **b)** Remove the side guards on the right-hand side. Slacken both take-up screws fully and note their positions.
- **c)** Remove the four bolts holding the right-hand return frame brackets and the bed plate retaining clips.
- **d)** Slide the old belt off sideways. Inspect the slider bed for wear and grooves while it is exposed.
- **e)** Check the direction arrow on the inside of the new belt (77-1012) and fit it with the arrow pointing towards the drive end on the top run.
- **f)** Refit the return frame brackets (tighten to 25 Nm), the bed plate clips and the guards.
- **g)** Tension the belt to 0.3 % (Section 6) and track it (Section 7).
- **h)** Re-check tension and tracking after 8 hours and 50 hours of running.

8.2 **Replacing a return idler**

Return idlers drop into slotted brackets on both side frames and are held by spring clips.

- **a)** Isolate and lock off the BCU-2. Slacken both take-up screws by two turns each to take the weight off the idlers.
- **b)** Remove the spring clips at both ends of the idler and lift it out of its slots.
- **c)** Fit the new idler (77-0410). Check that both hexagon shaft ends sit fully down in their slots, and refit the clips.
- **d)** Spin the idler by hand. It must turn freely and run true.
- **e)** Re-tension the belt by turning both screws back two turns, then check the tension marks (Section 6) and the tracking (Section 7).

8.3 Replacing the geared motor

The geared motor is mounted on the drive pulley shaft with a hollow shaft and torque arm.

a) Isolate and lock off. Unplug the motor cable at the motor.

b) Remove the torque arm bolt and the shaft end retaining bolt.

c) Draw the motor off the shaft with a puller on the extraction thread. Do not hammer on the gear housing.

d) Coat the shaft with anti-seize paste, slide the new motor on, and refit the retaining bolt (25 Nm) and the torque arm bolt with its rubber bush.

e) Check the direction of travel before running with product.

SECTION 9

Maintenance

Interval	Task
Daily	Clean the belt and pulleys of product residue. Check the belt runs centrally.
Weekly	Check belt edges for fraying. Check that the edge sensors are clean.
Monthly	Spin every return idler by hand. Check the belt tension marks. Check guard fixings.
6 months	Check frame squareness (4.2). Check pulley lagging for wear. Check the slider bed.
12 months	Check the gear unit for oil leaks (sealed for life, no oil change). Test the emergency stops.

9.1 Pulley and idler bearings are sealed for life and need no grease. Do not wash bearings with a high-pressure jet; it drives water past the seals.

9.2 Cleaning

Daily cleaning: isolate, scrape off loose product, wash the belt top and bottom with warm water and an approved detergent (2.5), rinse, and run the belt slowly for a few minutes with the guards on to dry it. Lift the bed plate clips once a week and clean under the slider bed, where product collects and lifts the belt.

9.3 Belt inspection

Inspect the belt surface for cuts, and the belt edges for fraying. An edge that frays on one side only is the record of a tracking problem: find the cause before fitting a new belt, or the new belt will go the same way.

SECTION 10

Fault finding

10.1 Fault codes shown on the BCU-2. Press RESET once the cause has been cleared.

Code	Fault	Likely cause	Action
C-01	Drive overload	Belt overloaded or jammed; motor thermistor trip; tension too high	Clear the belt. Check tension (6). Let the motor cool.
C-02	Belt mistrack	Belt edge over S3 (left) or S4 (right) for 2 s	Track the belt (7). Do not move the sensors.
C-03	Belt slip	Speed at S5 more than 5 % below drive speed for 5 s; low tension; wet or worn lagging	Check tension (6) and clean the drive pulley.

Code	Fault	Likely cause	Action
C-04	Emergency stop	E-stop or pull-wire operated	Find out why. Reset the E-stop, then RESET.
C-05	Product jam	Discharge photo-eye blocked longer than the set time	Clear the discharge. Check downstream equipment.
C-06	Safety relay fault	Safety relay or guard interlock fault (firmware 2.2 and later)	Check guard switches and wiring. Call Belton if it persists.

Symptom	Check
Belt runs to one side	Section 7.3, in order
Belt slips on start-up	Tension; drive pulley lagging; load on the belt at start
Squealing at the tail	Seized return idler or tail bearing
Belt edges fraying	Mistracking against the frame; edge sensor bracket bent

SECTION 11

Spare parts

11.1 Order parts from Belton Technical Support, quoting the serial number from the rating plate. Parts not listed here are specific to the conveyor length and are quoted by serial number.

Part no.	Description	Qty
77-1012	Belt, BX-12, 600 mm, PU food grade, endless (state conveyor length)	1
77-0410	Return idler roller, BX-12, complete with bearings	1 per metre

SECTION 12

Warranty

12.1 Belton warrants the BX-12 conveyor against defects in materials and workmanship for 12 months from the date of commissioning. The commissioning date is the date on the commissioning record returned to Belton (4.3). If no record has been returned, the warranty runs from the date of delivery.

12.2 The warranty does not cover belts, pulley lagging and slider beds, which wear in normal use, nor damage caused by overloading, incorrect tensioning, cleaning chemicals not approved by Belton, or parts not supplied by Belton.

12.3 To make a claim, contact Belton Technical Support with the serial number, the commissioning date, a description of the fault and, where possible, photographs. Do not return parts without a returns number.

SECTION 13

Declaration of conformity

13.1 Belton Conveyors Ltd declares that the BX-12 conveyor conforms to the Supply of Machinery (Safety) Regulations 2008 (UKCA) and to Machinery Directive 2006/42/EC (CE). Standards applied: BS EN 619:2002+A1:2010, BS EN ISO 12100:2010, BS EN 60204-1:2018. Signed for Belton Conveyors Ltd, Telford: R. Hollis, Engineering Director.

SECTION 14

Storage and disposal

14.1 Store the conveyor indoors, dry, with the belt tension released. A belt left tensioned for months takes a permanent set over the pulleys. Before returning to service, re-tension (Section 6) and track (Section 7) as for a new belt.

14.2 At the end of its life, the stainless steel frame, the aluminium motor housing and the copper windings can be recycled. The polyurethane belt and the BCU-2 electronics go to an authorised waste contractor under the WEEE Regulations.

SECTION

Commissioning record

Item	Entry
Serial number	
Site and line	
Conveyor length (centres)	
Frame diagonals (mm), left-right / right-left	
Belt elongation set (%)	
Tracking checked empty and loaded (Y/N)	
Edge sensor gap S3 / S4 (mm)	
BCU-2 firmware	
Date of commissioning	
Commissioned by (name, company)	
Signature	

SECTION

Maintenance log

Date	Hours	Work carried out / parts fitted	Name